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TRCT-GAN: CT reconstruction from biplane X-rays using transformer and generative adversarial networks

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ABSTRACT

Computed tomography (CT) provides a three-dimensional view of a patient's internal organs. Compared to CT volumes, X-ray imaging can significantly reduce the patient's exposure to ionizing radiation. Moreover, X-ray images are more economical and widely applied in surgical procedures. However, X-ray images can only provide two-dimensional information. In this paper, an end-to-end GAN network, named TRCT-GAN, is proposed to reconstruct chest CT volumes from biplane X-ray images. In the GAN network, the Transformer network module is employed to enhance the feature representation of X-ray images. Moreover, a dynamic attention module is added to exploit some 2D feature maps and 3D feature maps to enhance the contextual association. The experimental results demonstrate that the proposed network can effectively produce high-quality CT reconstructions from X-ray images.

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1. Introduction

X-rays are an imaging technique that has been widely applied in the medical field. It allows us to observe the body inside non-invasively, such as tissues, organs, bones, etc. However, these tissues and organs are projected on a two-dimensional plane, in which some information is missed. Computed tomography (CT) solves this problem by displaying the tissues, organs, and bones in three dimensions. In a CT scan, for a better understanding of the patient's injuries, the X-ray equipment rotates around the patient's body to examine their internal anatomy in three dimensions. Therefore, CT images provide clearer anatomical and spatial information than 2D images [1,2]. Nevertheless, for CT scans, the patients may suffer from excessive radiation. As a result, the risk of cancer is increased significantly compared to X-rays with low radiation. More, the equipment of CT is expensive and bulky. It may be not available to perform a CT scan in some backward areas. For general hospitals, two-dimensional X-ray image acquisition equipment is widely used in surgical procedures because of the low radiation dose and the low price.

However, the high-resolution 3D images are generally more helpful in actual treatment, compared to low-dimensional and local images, such as endoscopic and X-ray images. In order to deal with this problem, 3D anatomical shape reconstruction based on 2D X-ray images has been gradually developed [3–6] and implemented in various parts of the human body or animals. As one kind of effective approach, many end-to-end supervised deep neural networks have been reported to reconstruct CT [7,8]. In [9], a study has been conducted to perform CT reconstruction of X-rays to animal skulls by using deep learning CNN networks. It is a single-view reconstruction of the CT, encoded and decoded using a conventional CNN network. There is a hop-connected network between the encoder and the decoder. Reconstruction of the three-dimensional structure of teeth from two-dimensional X-ray images was achieved by Weinan Song et al. [10]. In the CT reconstruction of the teeth, the author designed a GAN network. In its generator part, a densely connected network was used to extract the features. Similarly, a hop-connected network was also used to connect the encoder and decoder. Yoni Kasten et al. proposed an end-to-end deep learning network for 3D reconstruction of knee bones [11]. It is still mainly the jump connection network that is used in the reconstruction of the knee CT. Zhiqiang Tan et al. proposed a lightweight CNN multiscale feature fusion network to reconstruct lung CT [12]. To further improve the quality of the reconstructed CTs, Zhiqiang Tan et al. also added an attention module to the network. Rongjun Ge et al. also proposed a network for reconstructing cervical spine CT [13], decomposing features from overlapping domains to corresponding stereoscopic locations from

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multiple scales. A network that optimizes the reconstructed CT is added later. Xueqin Sun et al. used a densely connected network for encoding, with each layer of the encoder and decoder interconnected, to achieve an X-ray picture-to-CT mapping for analyzing the combustion surface of a solid rocket motor (SRM) [14]. An end-to-end reconstruction method of lung CT was studied by Liyue Shen et al. [15,16], in which multiple input X-ray images are tried for CT reconstruction.

Deep learning methods are used to learn the mapping from 2D X-ray images to 3D CT volumes. Although a single X-ray image can also reconstruct the CT volume, dual-plane X-ray images can reconstruct a higher-quality CT volume. Recently, generative adversarial networks (GAN) [17] have been widely used in medical image processing [18][19][20]. A GAN-based network framework called X2CT-GAN [21] has been implemented for both CT reconstruction from a single X-ray image and CT volume reconstruction from biplane X-ray images by learning the anatomy of the chest.

Nowadays, there are still some intractable problems for CT reconstruction, e.g., the reconstructed large organs are indistinct, the lung field and small bronchi are not clear, etc. In our work, based on the X2CT-GAN network, an effective CT reconstruction network is devised by utilizing a TRCT-GAN module, a 2D Attention In Attention (2D AIA), a Transformer module, a 3D AIA module, and a perceptual loss function. The main contributions of the proposed work can be summarized as follows: (1) Due to the complex structure of X-ray images, the proposed network can utilize remote contextual information well. Inspired by [22], we propose a 2D AIA module to have better feature representation for encoding. More useful features are extracted in the encoder part, which solves the problem that the boundaries are not clear enough when we reconstruct larger organs and structures. (2) In the reconstruction of 2D X-ray images to 3D CT volumes, the most important thing is the conversion of 2D features to 3D features. In the proposed work, the Transformer module [23] is employed to perform the feature transformation in order to establish a better association between 2D features and 3D features. It also reduces the loss of some useful features when connecting 2D and 3D features. (3) In the decoding part, the number of parameters is large because of 3D upsampling. We propose a 3D Attention In Attention (3D AIA) module based on 2D AIA. There are two upsampling methods in the 3D AIA module, one is the nearest neighbor interpolation algorithm and the other is the trilinear interpolation algorithm. The 3D AIA module achieves a balance of performance by enhancing the model reconstruction with a small number of additional parameters. After adding 3D AIA, when we fuse dual-view features later, we can fuse more useful 3D features together and reconstruct CT with better visual effects. The reconstructed CT has a complex structure of the surface and interior. In the proposed work, a perceptual loss function [24] is adopted to constrain it to accurately reconstruct the structure and texture of CT.

2. Materials and methods

To reconstruct the CT volume using both frontal X-rays images and side synthetic X-rays, we used two parallel encoder-decoder networks and added a fusion network at the decoder to fuse the dual view information. Compared to traditional CNN networks, the visualization of the GAN network is better. In our network, several improvements have been made for X2CT, 2D dynamic attention, and 3D dynamic attention are added in the low channel of the encoder and decoder respectively, the Transformer module is cascaded in the 2D to 3D mapping. In order to achieve better performance, we added perceptual loss functions to constrain reconstruction again.

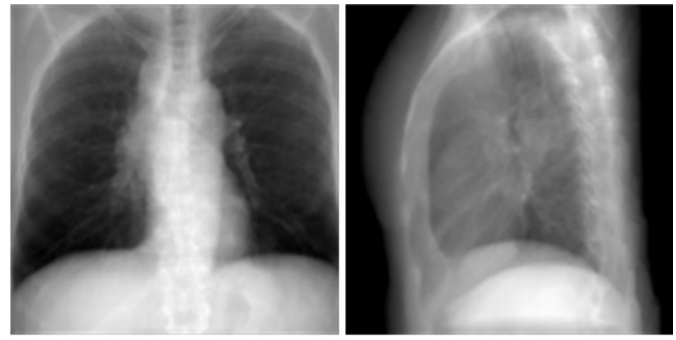

Fig. 1. DRRs generated from real CT volumes.

2.1. Dataset

For training, we need datasets containing multiple paired X-ray photographs and CTs, but it is too difficult to obtain such a large number of data. In many existing works on lung reconstruction, CT volumes were only projected onto a 2D plane to obtain digitally reconstructed radiographs (DRRs) [25] in the place of X-ray images [15,16]. The reason is that it is difficult to obtain annotated X-ray images and corresponding CT scan datasets. Moreover, it is also unethical and unkind to expose patients to excessive radiation.

As shown in Fig. 1, the corresponding X-ray images are synthesized using the DRRs technique. We define an X-ray source with virtual X-rays that are passed through the CT and projected toward the backplane. In this process, the intersection of the point source with each CT slice in the CT body data can be calculated to obtain the corresponding CT value. After a ray projection is finished, the CT value obtained by accumulating the whole path is the corresponding DRR image pixel value. Finally, the CT value of each pixel block of the whole DRR image will be obtained, and the totalized value will be mapped to the pixel gray value to obtain a DRR image. A well-known public lung dataset, LIDC-IDRI [26], with 1018 scans of CTs, is adopted in our work. In the dataset, there are synthesized 1018 frontal and lateral X-ray images. Among these, 916 X-ray images are used as the training set, and the rest of the data is used as the test set. Before input to the network, the size of CT volumes and X-ray images were adjusted to $128\times128\times128$ and 128×128, respectively.

2.2. Network architecture

Similar to other 3D GANs, the network structure includes a 3D generator and a 3D discriminator. As shown in Fig. 2, in the generator part, two parallel encoder-decoders take the front and side synthetic X-ray photos as input to the network, respectively. A jump connection is added between the encoder and decoder. Moreover, a fusion network is adopted to fuse the frontal and lateral information to synthesize the 3D CT.

Due to the complex X-ray structure, a densely connected module [27] is used in the encoder part to extract features. Furthermore, dynamic attention is added in the encoder part to optimize the feature extraction, as shown in Fig. 3. When the numbers of channels are 64 and 128, the input features are passed through the 2D AIA before entering the dense connection module. The 2D AIA module consists of a non-attention branch coupled with an attention branch, and a third branch to generate dynamic attention weights for the first two branches. The attention branch is used to enhance the information that is valuable to the network, and the non-attention branch is used to learn information that is ignored by the attention branch. 2D AIA networks can control the attention module to focus more on valuable features. Moreover, the capacity

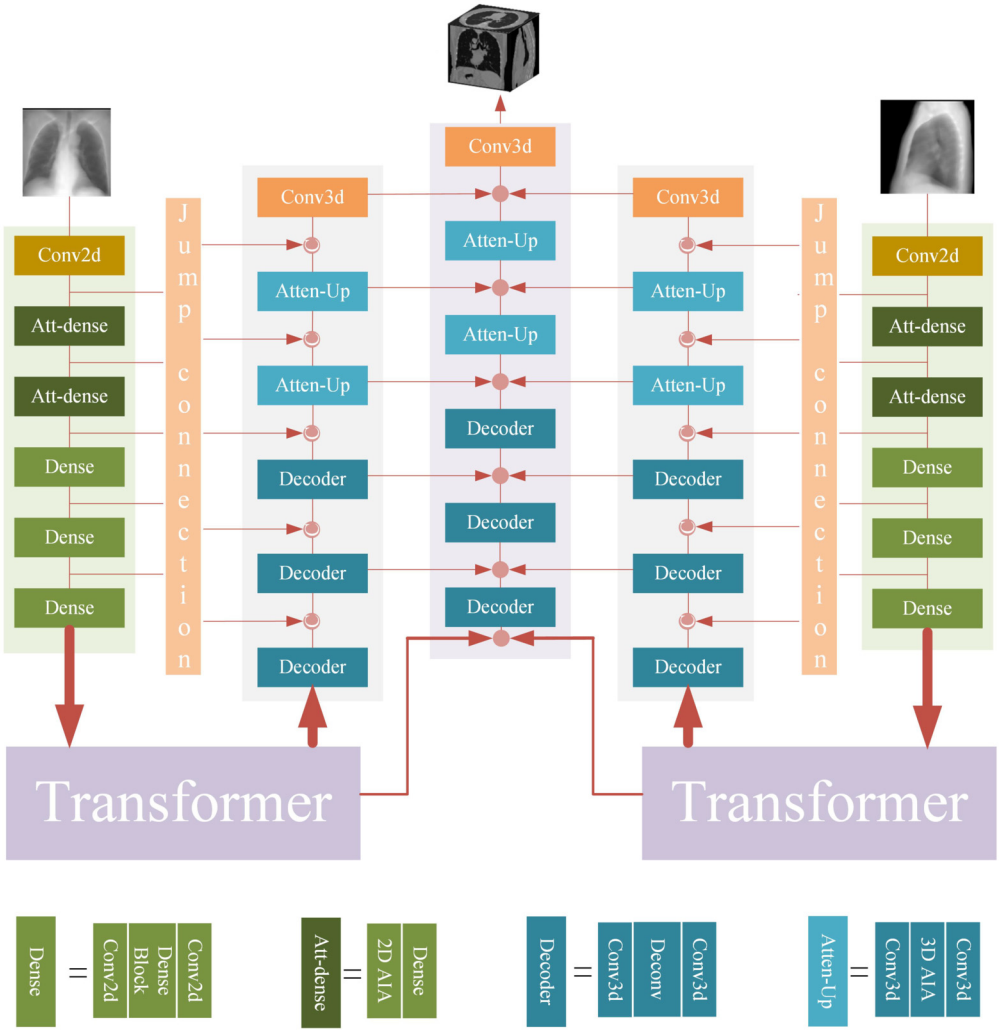

Fig. 2. The generator network structure of TRCT-GAN.

of the attention module can be greatly increased while reducing parameters. We can improve our ability to extract useful features with a small number of parameters, making our reconstructed lung CT organ boundaries clearer. These can be formulated as follows:

$$y = D_a(x) A_2(x) + D_b(x) B_2(x) \tag{1}$$

where x is the input feature and y is the output feature. $A_2(x)$ is the output of a two-dimensional attention branch and $B_2(x)$ is the output of a two-dimensional non-attention branch. $D_a(x)$ and $D_b(x)$ are the weights of the attention and non-attention branches, respectively.

If the mapping from 2D to 3D is achieved by simple convolution, it may cause 2D spatial feature loss and affect the final 3D reconstruction effect. Therefore, to reduce 2D spatial feature loss, a Transformer module is added to bridge 2D features and 3D features. One main advantage of the Transformer is it's global computational power. Compared to CNN, the Transformer module can retain more valuable spatial information because it can increase the image's receptive area, gather more contextual data, and capture the rich feature information of X-ray images. It is crucial in transforming 2D X-ray image data into 3D CT spatial features. As shown in Fig. 4, the 2D features are turned into 1D features. After embedding and positional encoding, they are directly summed up and used as the input of the Transformer. The input vector goes through a Layernorm layer, and then through a multi-head self-attention layer to make a residual connection. After the Layernorm layer and the Feed Forward layer, it is connected again. This reduces the loss of 2D features to 3D feature mapping and greatly improves the final 3D CT reconstruction.

The upsampling is first done in the decoder by 3D convolution and 3D deconvolution. However, in the last two lower channels, such an upsampling method cannot fully utilize the 3D features mapped over. Therefore, in the last two low channels, we add a 3D AIA network. When the features of the two X-ray images are complex, it can well fuse the dual-view 3D features. The 3D AIA network can pay more attention to the fusion of useful 3D features in the two views, and reconstruct CT with better visual effects. As shown in Fig. 5, it is a 3D dynamic attention network. In the 3D AIA module, we use two upsampling algorithms, the trilinear interpolation algorithm [28] and the nearest neighbor interpolation algorithm [29]. The nearest-neighbor interpolation algorithm can sample fast because of less computation burden. However, it will lead the quality degradation due to not considering the values of other surrounding voxel points. Compared with the nearest-neighbor interpolation algorithm, the sampling speed of the trilinear interpolation algorithm is slightly slower because of its high computational intensity. However, the trilinear interpolation algorithm has a better interpolation image. Therefore, a combination of them is adopted in our work to achieve a balanced effect. These can be formulated as follows:

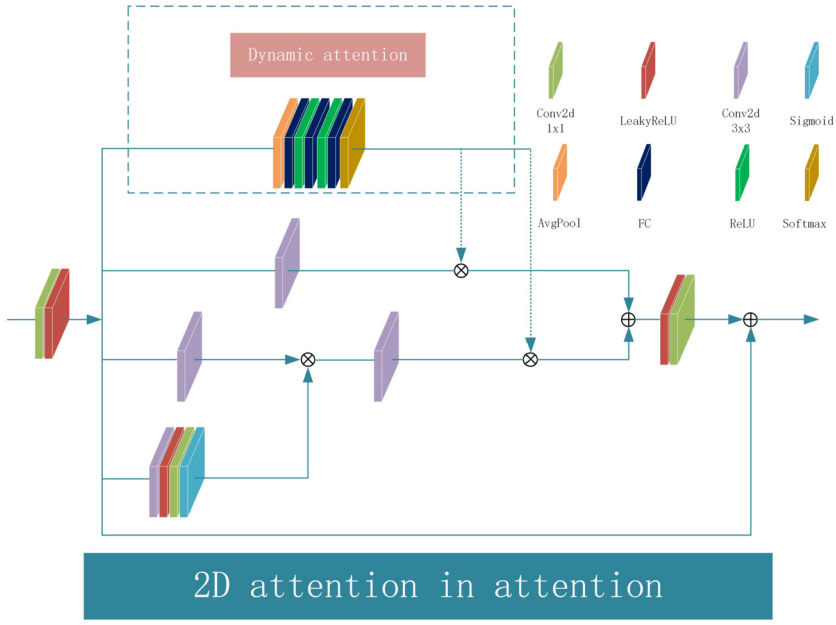

Fig. 3. The two-dimensional attention in attention module (2D AIA).

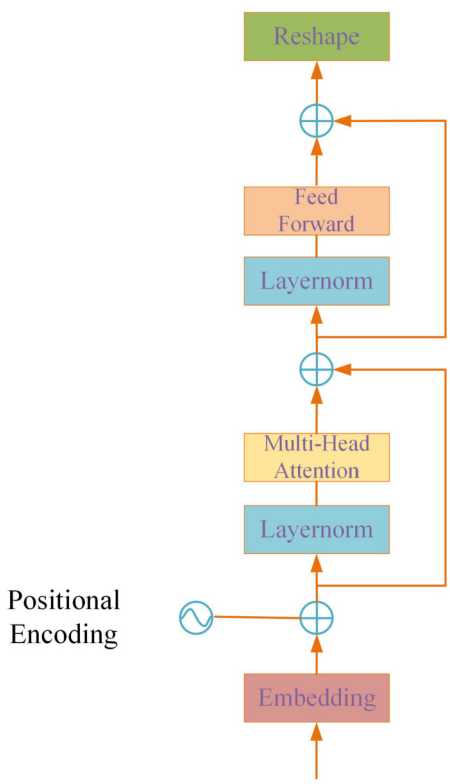

Fig. 4. One of the layers in the Transformer module.

$$I = f_n(t_d) + f_t(t) \tag{2}$$

where I is the output 3D feature, t_d is the output feature that has gone through the dynamic attention module, t is the feature that has not gone through the dynamic attention module, $f_n(\cdot)$ is the nearest neighbor interpolation function, and $f_t(\cdot)$ is the trilinear interpolation function.

Considering information complementarity, the jump connection, and the feature fusion are respectively applied between the encoder and decoder and between two views to reconstruct a fine model. The jump connection module is shown in Fig. 6. A third decoder network is used to fuse the extracted two-view information and output the reconstructed CT volume.

A widely used discriminator part, named PatchGAN [30–34], is also adopted in our work. The difference from the normal GAN network discriminator is that PatchGAN maps inputs into matrices and then averages them into a final output. Therefore, PatchGAN has good generalization performance by taking into account the influence of different parts of the feature map. The discriminator consists of three parts, i.e. the 3D convolution, the instance normalization layer [35], and the activation function.

2.3. Loss function

In the proposed network, we use several loss functions for constraining the generative model, namely, adversarial loss, reconstruction loss, projection loss, and perceptual loss function. Similar to [21], the weights are 0.1, 10, 10, and 7, respectively.

Among the adversarial loss function, we use the loss function of LSGAN [36] with the following expressions:

$$\mathcal{L}(D) = \frac{1}{2}\Big[\mathbb{E}_{y\sim p(CT)}\left(D(y|x)-1\right)^2 + \mathbb{E}_{x\sim p(X-rays)}\left(D(G(x)|x)-0\right)^2\Big] \tag{3}$$

$$\mathcal{L}(G) = \frac{1}{2}\Big[\mathbb{E}_{y\sim p(X-rays)}\left(D(G(x)|x)-1\right)^2\Big] \tag{4}$$

where x is the input two synthetic X-rays and y is the reconstructed CT. Discriminator D and generator G are alternately trained to compete with each other.

Reconstruction loss is a loss of voxels in 3D space that enhances the similarity of the reconstructed CT spatial structure. The projection loss is projected from three angles separately using orthogonal projection. The reconstruction loss and projection loss are defined as follows.

$$\mathcal{L}_{re} = \mathbb{E}\|y - G(x)\|_2^2 \tag{5}$$

$$\mathcal{L}_p = \frac{1}{3}[\mathbb{E}_{x,y}\|P_{ap}(y) - P_{ap}(G(x))\|_1 + \mathbb{E}_{x,y}\|P_{cp}(y) - P_{cp}(G(x))\|_1 + \mathbb{E}_{x,y}\|P_{sp}(y) - P_{sp}(G(x))\|_1] \tag{6}$$

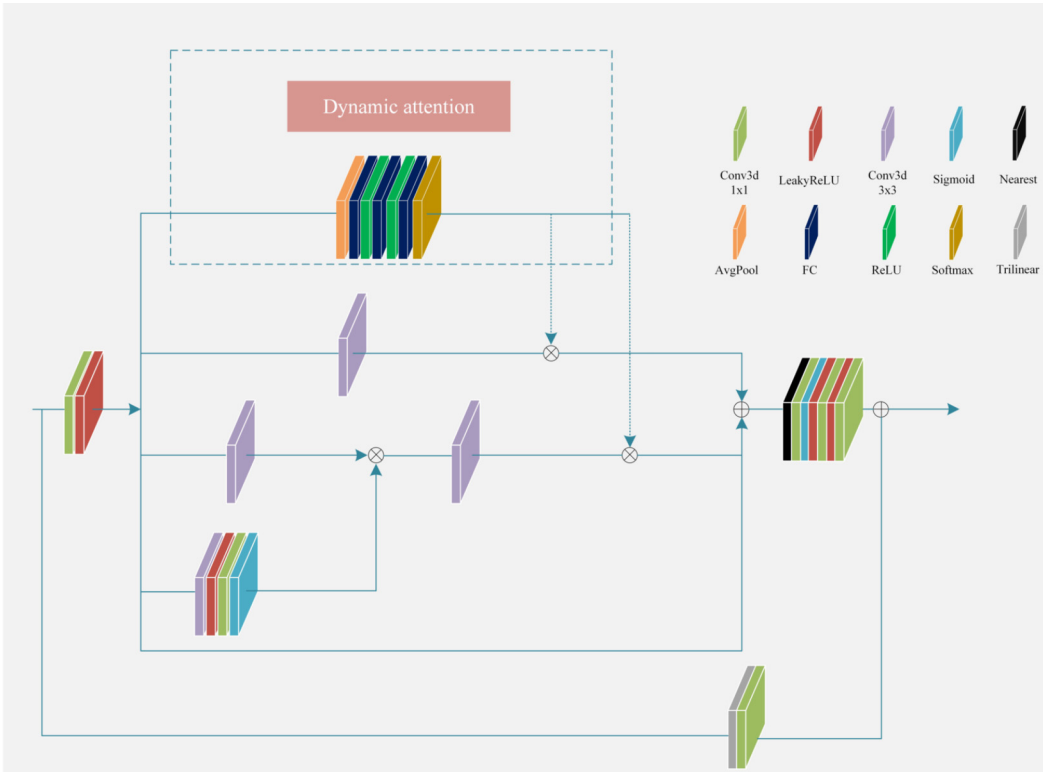

Fig. 5. 3D AIA network. The Nearest module is the nearest neighbor interpolation algorithm and the Trilinear module is the trilinear interpolation algorithm.

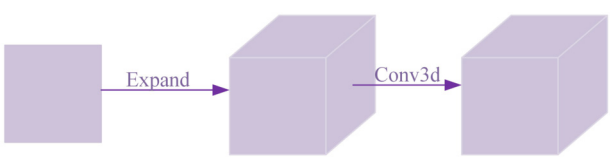

Fig. 6. The Jump connection module.

where $P_{ap}(\cdot)$ represents the projection on the axial plane, $P_{cp}(\cdot)$ represents the projection on the coronal plane, and $P_{sp}(\cdot)$ represents the projection on the sagittal plane.

Since orthogonal projection causes some information loss, we use the perceptual loss function to optimize the projection loss function. The perceptual loss function is used to compare the differences in style and content between two-dimensional images, and thus reduce the differences between them. So we first just project the real CT and the reconstructed CT on the two-dimensional plane from three angles to get the two-dimensional projection images. Then we use the perceptual loss function to compare the differences between the corresponding projection images and reduce the differences between the real CT projection images and the reconstructed CT projection images. Eventually, the difference between the reconstructed CT and the real CT is also reduced accordingly. Thus, a better reconstruction effect can be produced. The specific operation is as follows. We project the reconstructed CT and the real CT onto a two-dimensional plane. The size of both obtained 2D images is 128×128. Up-sampling the two image sizes to 224×224 is used as the input to the VGG16 network. Finally, the features obtained by convolving the real 2D planar image are compared with the features obtained by convolving the generated 2D planar image so that the content and global structure of both are close. The expression is as follows.

$$\begin{aligned}\mathcal{L}_{per} = \frac{1}{3}[&\mathbb{E}_{x,y}\|P_{ap}(V(y)) - P_{ap}(V(G(x)))\|_2^2 + \\ &\mathbb{E}_{x,y}\|P_{cp}(V(y)) - P_{cp}(V(G(x)))\|_2^2 + \\ &\mathbb{E}_{x,y}\|P_{sp}(V(y)) - P_{sp}(V(G(x)))\|_2^2]\end{aligned} \tag{7}$$

where V indicates that the input has been pre-trained to the model by VGG [37].

3. Results

In this section, some qualitative and quantitative analyses are performed to demonstrate the validity of our proposed TRCT-GAN network. We also display the network settings and ablation study.

3.1. Network settings

In the training, the Adam optimizer [38] is used to train the network 100 times with an initial learning rate of $4e{-}4$ and momentum parameters β_1=0.5,β_2=0.99. A total of 100 training steps were performed. After 70 steps, the learning rate was reduced to 0 in the last step using a linear learning rate decay strategy. In our experiments, we use instance normalization [35] instead of using batch normalization [39]. Training and testing were run on an NVIDIA TESLA V100S with 32g of video memory.

3.2. Qualitative analysis

The evaluation results of the qualitative analysis are displayed in Fig. 7. Note that '+B' represents the two-plane X-ray input and '+S' means the single-view X-ray input. Fig. 7 shows that both TRCT-GAN and X2CT-GAN networks have better visualization than their corresponding CNN networks. Besides, by comparing their axial views, the reconstruction of TRCT-GAN+B is much more accurate than that of X2CT-GAN+B, especially the marked ones, which retain more anatomical details. Specifically, the CT reconstructed

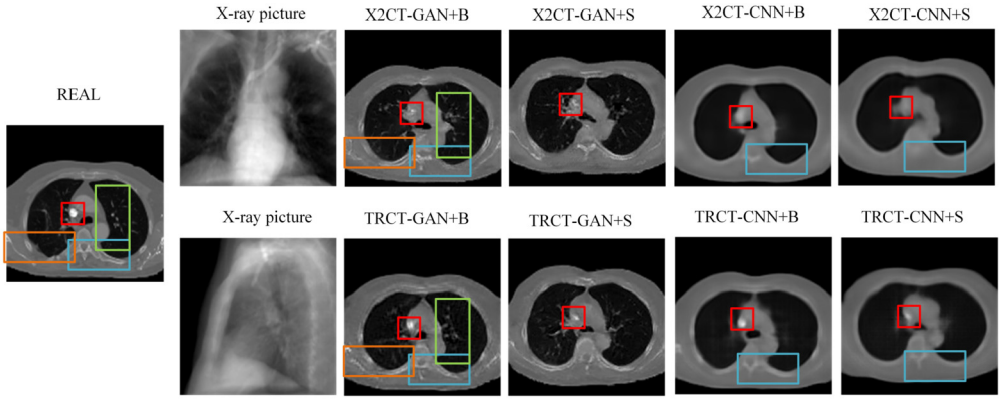

Fig. 7. The leftmost is the axial view of the real CT volume. Except for our composite X-ray image. The first row is the axial attempt of the CT volume generated by X2CT-GAN+B, X2CT-GAN+S, X2CT-CNN+B, and X2CT-CNN+S. The second row is the axial attempts generated by TRCT-GAN+B, TRCT-GAN+S, TRCT-CNN+B, and TRCT-CNN+S. The TRCT network can generate more accurate anatomical reconstructions than X2CT.

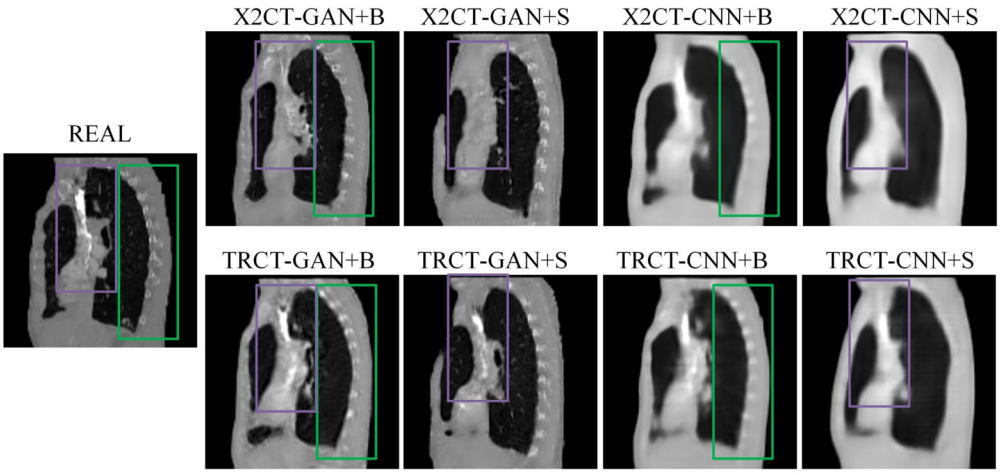

Fig. 8. The leftmost one is the side view of the real CT volume. The first row is the side view of the CT volume generated by X2CT-GAN+B, X2CT-GAN+S, X2CT-CNN+B, and X2CT-CNN+S. The second row is the side view generated by TRCT-GAN+B, TRCT-GAN+S, TRCT-CNN+B, and TRCT-CNN+S. The TRCT network can produce more accurate anatomical reconstruction than X2CT.

by our method shows more accurate tissue morphology and density compared to that reconstructed by the X2CT-GAN+B network. The size of the solid diameter of the high-density lesion plaques is more comprehensive and well-defined compared to that of the X2CT-GAN+B network. The rib and vertebral transverse processes show that more uniform bone density, and the extent of the lung field and the course of the small bronchi are more clearly defined. In any case, our reconstructed CT cannot replace the real CT due to the ultra-low radiation of our X-ray images. However, as shown in the figure, the CT reconstructed from the X-ray images is still superior to the original X-ray images, especially by solving its problem of overlapping tissues and organs and showing some of the tissues better. This is a good complement to the X-ray image alone. Furthermore, we can see that the X2CT-GAN, X2CT-CNN, and TRCT-GAN, TRCT-CNN networks reconstructed CTs with better visual results when their input was a biplane X-ray image compared to a single X-ray image as the input to the network. This confirms that CT volume reconstruction using dual-view x-rays has better results compared to using single-view rays.

Simultaneously, to show the reconstruction effect from the side view, the CT volume reconstruction results of the TRCT and X2CT network are illustrated in Fig. 8, with a single-view X-ray and a double-view X-ray as the input respectively. In summary, the reconstruction performance is excellent in both axial and lateral views as well as in anatomical structure and shape. In addition, as shown in Fig. 9, it can also be seen that our method accurately reconstructs the main anatomical structures and shapes of the CT volume.

3.3. Quantitative analysis

To quantitatively analyze the reconstruction performance of our proposed method and others, two widely used metrics are employed, including peak signal-to-noise ratio (PSNR) and structural similarity (SSIM) index. PSNR is usually used to measure the quality of reconstructed digital signals, and SSIM is an important metric to judge the similarity (brightness, contrast, structure) of two images.

We summarize the quantitative results in Table 1. First, the PSNR and SSIM metrics of the CTs generated by the 2DCCN network were compared. It can be found that the PSNR and SSIM of the CT reconstructed by all our CNN methods are larger than that of the CT reconstructed by 2DCNN. Compared to single-plane X-rays as the input, both PSNR and SSIM are significantly improved in TRCT-GAN and TRCT-CNN with biplane rays as the input to the network. As can be seen in Table 1, the PSNR and SSIM of TRCT-GAN+B are 4.5 dB and 0.162 higher than those of TRCT-GAN+S, respectively. The PSNR and SSIM of TRCT-CNN+B are 4.25 dB and 0.135, which are higher than that of TRCT-CNN+S. The PSNR and SSIM of X2CT-GAN+B are 3.89 dB and 0.131 higher than those of X2CT-GAN+S, respectively. The PSNR and SSIM of X2CT-CNN+B are 4.17 dB and 0.134 higher than X2CT-CNN+S respectively. In terms

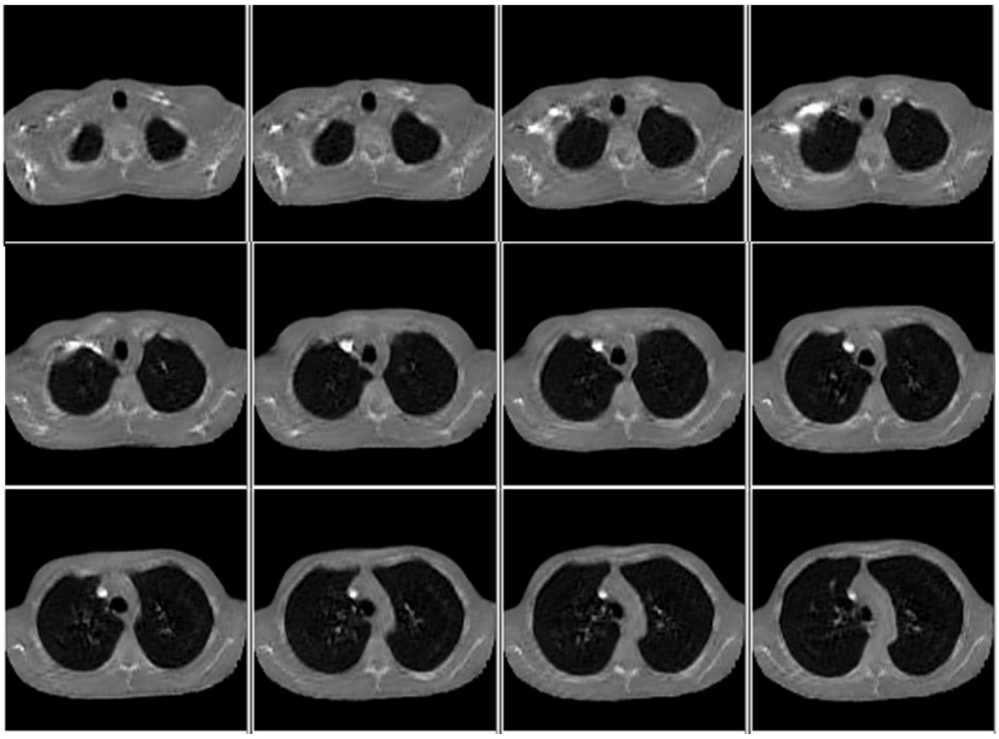

Fig. 9. Slice image of reconstructed CT volume.

Table 1
Quantitative results from TRCT, X2CT and 2DCNN [40]. TRCT is the proposed generator network, and 'TRCT-GAN' is our GAN-based method. Here, '+S' denotes the single-view X-ray input, and '+B' indicates the biplanar X-ray input. Better results are in bold.

Network	PSNR (dB)	SSIM
2DCNN	23.10	0.461
X2CT-GAN+B	26.19	0.656
TRCT-GAN+B	**27.01**	**0.705**
X2CT-GAN+S	22.30	0.525
TRCT-GAN+S	**22.51**	**0.543**
X2CT-CNN+B	27.29	0.721
TRCT-CNN+B	**27.46**	**0.726**
X2CT-CNN+S	23.12	0.587
TRCT-CNN+S	**23.21**	**0.591**

of metrics, it is once again proven that a biplane X-ray image is better as an input than a single X-ray image as an input. Since the quality of the CT volumetric visual images generated by the GAN network is better than that generated by the CNN network in the qualitative results, we focus on comparing the TRCT-GAN network with the X2XT-GAN network. When the two-view X-ray is used as the input, the PSNR and SSIM of CT volume generated by the TRCT-GAN+B network are respectively 0.82 dB and 0.049 high than those generated by our baseline network X2CT-GAN+B. The PSNR and SSIM of the CT volume generated by the TRCT-GAN+S network are 0.21 dB and 0.018 higher than those of the CT volume generated by our baseline network X2CT-GAN+S, respectively. To demonstrate the quality of our reconstructed CT volume, we make a more detailed comparison in Table 2, by calculating the PSNR and SSIM of the 2D projection plane of the reconstructed CT from three more views in the coronal plane sagittal plane transverse plane, respectively. The results in Tables 1 and 2 are all average values of the PSNR indicator and SSIM indicator for the test data. Therefore, a box plot of the indicators is also provided here, as shown in Fig. 10.

3.4. Ablation study

To verify the effectiveness of the extra perceptual loss function, the Transformer module, 2D dynamic attention module, and 3D dynamic attention module, we conducted an ablation study about TRCT-GAN, as shown in Table 3.

4. Discussion

In the qualitative analysis, our method has improved PSNR and SSIM in both the GAN network framework and CNN network framework compared to previous studies. The quantitative analysis similarly demonstrated that our method in this paper outperformed previous studies with the same dataset. To demonstrate the value of this study, we reconstructed the lung CT volumes using a pair of real biplane X-ray images of the lungs as the input to the network. As shown in Fig. 11 and Fig. 12, it is clear that the real X-ray images are sharper than the synthetic X-ray images, and the visual quality of the reconstructed CT volumes using the real biplane X-ray images is also high. Compared to the original real 2D X-ray images, the reconstructed 3D CT volume allows the physician to visualize more anatomical details and provides more spatial information to the physician.

In the field of CT reconstruction, previous research work has been devoted to CT reconstruction of teeth, spine, skull, kneecap, liver, and chest. Compared to CT reconstruction of bones and teeth, CT reconstruction of the chest is more difficult because the chest has bones, organs, and soft tissues. In addition, the dataset of chest CT used in this paper is a bit small when a large number of datasets can optimize the performance of network reconstruction. Therefore, in order to reconstruct an accurate CT with a small amount of data, the number of network parameters proposed in this paper is high and the computational cost is high. The purpose of the present work is not to use the CT reconstructed from X-ray images to replace the original real CT. Because although our proposed method can reconstruct larger organs, bones, and tissue structures, it is still difficult to reconstruct some of the fine

Table 2
PSNR-1, PSNR-2, and PSNR-3 are PSNR indicators obtained from CT coronal, sagittal and transverse views. SSIM-1, SSIM-2, and SSIM-3 are SSIM indicators obtained from CT coronal, sagittal and transverse views.

Network	PSNR-1 (dB)	PSNR-2 (dB)	PSNR-3 (dB)	SSIM-1	SSIM-2	SSIM-3
X2CT-GAN+B	28.79	35.29	26.95	0.651	0.666	0.649
TRCT-GAN+B	**30.09**	**37.46**	**27.83**	**0.701**	**0.715**	**0.699**

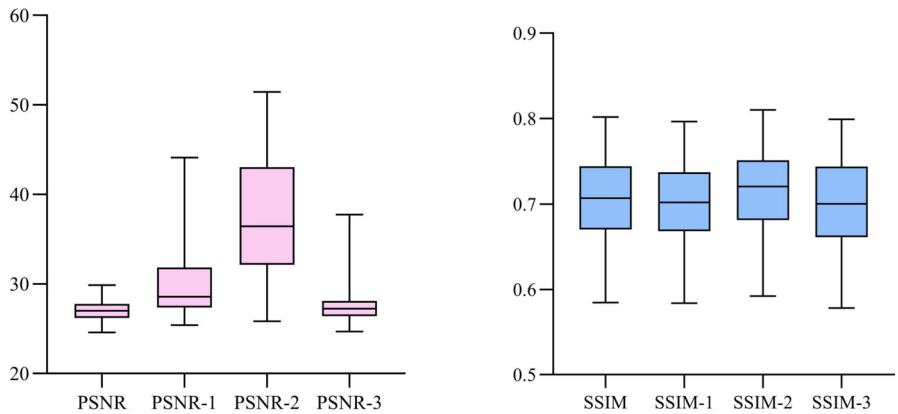

Fig. 10. Box plots of the PSNR and SSIM metrics for CT generated by the TRCT-GAN+B network.

Table 3
Several combinations based on TRCT-GAN+B and the best results are bolded.

Combination				Metrics	
PerLoss	Transformer	2D AIA	3D AIA	PSNR (dB)	SSIM
				26.19	0.656
✓				26.37	0.675
✓	✓			26.49	0.684
✓	✓	✓		26.65	0.693
✓	✓	✓	✓	**27.01**	**0.705**

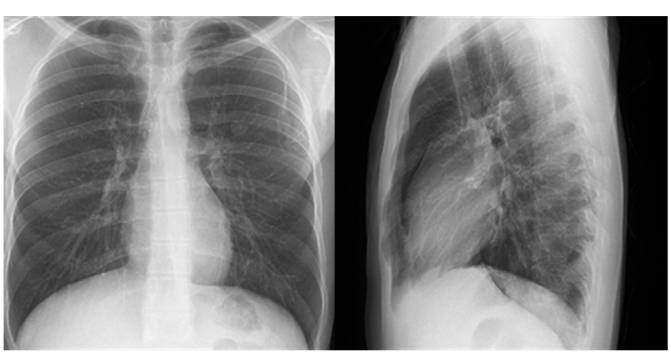

Fig. 11. Realistic biplane x-ray images.

anatomical structures and the reconstructed images have some artifacts. However, compared to our original input X-ray images, we can separate the overlapping organs, bones, and tissue structures in the 2D plane in 3D space to obtain our CT slices. It is more beneficial to the doctor's diagnosis and observation. The reconstructed CT has great value and it is a good complementary enhancement to the low-cost X-ray machine because when we can only get X-ray images of the patient because of equipment or time, we can reconstruct the CT volume with X-ray images to give the physician more information.

5. Conclusion

In this paper, we propose a TRCT-GAN network for reconstructing 3D CT volumes from X-ray images. Compared to the existing approaches, Experimental results demonstrate that our method can reconstruct CT volumes with higher visual quality and higher quality anatomical structures and shapes.

CRediT authorship contribution statement

Yufeng Wang: Formal analysis, Methodology, Validation, Visualization, Writing – original draft. **Zhan-Li Sun:** Formal analysis, Resources, Supervision, Writing – review & editing. **Zhigang Zeng:** Resources, Supervision, Writing – review & editing. **Kin-Man Lam:** Resources, Supervision, Writing – review & editing.

Declaration of competing interest

We declare that we have no financial and personal relationships with other people or organizations that can inappropriately influence our work, there is no professional or other personal interest of any nature or kind in any product, service and/or company that could be construed as influencing the position presented in, or the review of, the manuscript entitled, "TRCT-GAN: CT Reconstruction from Biplane X-Rays using Transformer and Generative Adversarial Networks".

Data availability

The data that has been used is confidential.

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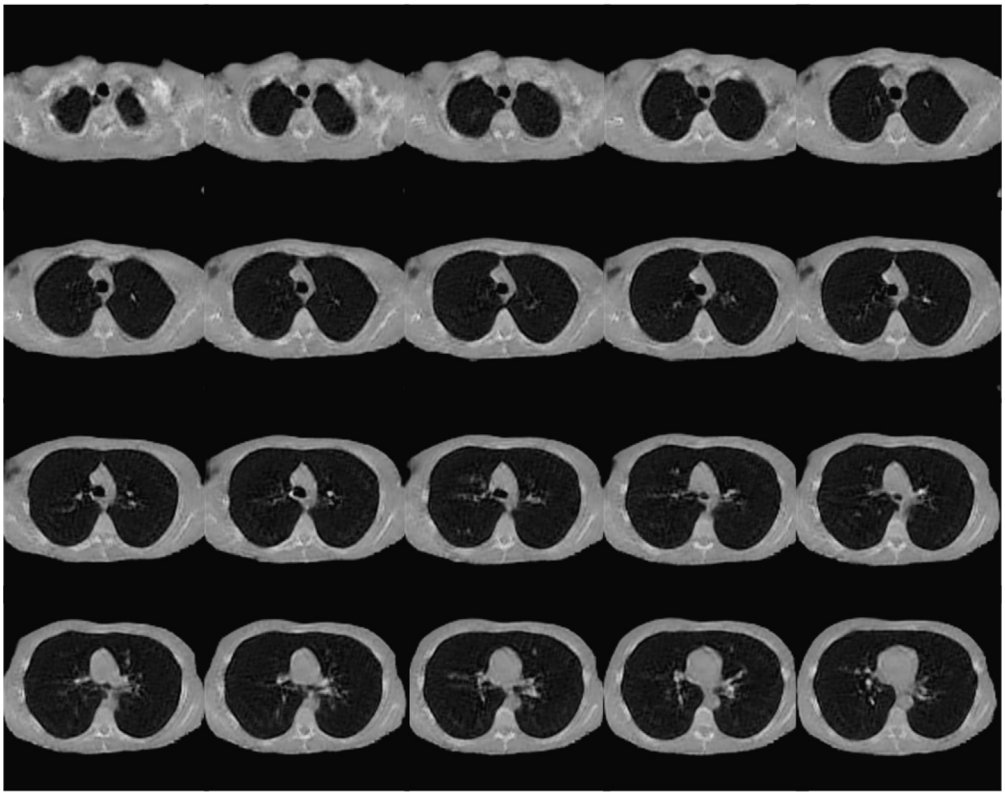

Fig. 12. CT volume reconstructed from real X-ray images.

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